

# PRIYA VENKATESH

Software Engineering & Cloud Systems Intern

## CONTACT & LINKS


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Hyderabad, India

LinkedIn

GitHub

Portfolio


## SKILLS

### TECHNICAL

Java · Spring Boot · Python · FastAPI · React · TypeScript · Kubernetes · Redis · PostgreSQL · Docker

### SOFT SKILLS

Agile / Scrum · Cross-Functional Collaboration · System Architecture

## CERTIFICATIONS

**Kubernetes and Cloud Native Associate (KCNA)**
Linux Foundation & CNCF · 2024

**Oracle Certified Professional: Java SE 11 Developer**
Oracle · 2023

## ABOUT ME

Final-year Computer Science student with practical production experience building microservices and cloud automation pipelines. Proven ability to reduce query latency and automate CI/CD deployments.

## EDUCATION

**Indian Institute of Information Technology** — 2021 – 2025
B.Tech in Information Technology in Information Technology — **Score: 9.15 / 10.0 CGPA**

**Delhi Public School** — 2019 – 2021
Higher Secondary Certificate (CBSE Class XII) in Physics, Chemistry, Mathematics — **Score: 96.4%**

## PROJECTS

### Distributed Distributed Task Queue

Lead Architect

Engineered a fault-tolerant asynchronous job queue with dead-letter exchanges and exponential backoff retry policies.

- Benchmarked task processing throughput exceeding 12,000 tasks/second with sub-5ms Redis latency.
- Designed graceful worker shutdown ensuring zero in-flight task loss during rolling cluster updates.
- Built Prometheus metrics exporter monitoring queue lag, worker memory, and failure rates.

*Technologies: Go, Redis, PostgreSQL, Docker*

Live Demo · Source Code

### Real-Time Collaborative Code Editor

Frontend & WebSocket Developer

Developed an in-browser multiplayer code editor with operational transformation and multi-cursor synchronization.

- Implemented Yjs CRDT protocol resolving concurrent keystrokes across geographically distributed clients.
- Added syntax highlighting for 14 programming languages with Monaco Editor integration.
- Containerized code execution sandbox executing Python, C++, and JavaScript with strict time limits.

*Technologies: TypeScript, React, Node.js, WebSocket, CRDT*

Live Demo · Source Code

### Smart Campus Shuttle Tracker

Mobile & Backend Developer

GPS telemetry platform tracking 12 campus transit shuttles in real time for 8,500 daily university commuters.

- Streamed vehicle coordinates over MQTT with sub-second websocket client broadcast.
- Reduced student wait times by an estimated 18 minutes through predictive arrival algorithms.

*Technologies: FastAPI, PostgreSQL, Leaflet.js, MQTT*

Source Code

## EXPERIENCE

**Backend Software Engineering Intern — Razorpay Technologies** May 2024 – Jul 2024
Bangalore, India

Worked with the Merchant Settlements team optimizing payment reconciliation batches and idempotency safeguards.

- Refactored batch settlement SQL queries reducing nightly execution window from 42 minutes to 16 minutes.
- Implemented Redis distributed locks preventing duplicate payout dispatches under concurrent webhooks.
- Wrote 45+ unit and integration tests achieving 91% code coverage on settlement reconciliation modules.

**Student Research Assistant — Cloud Computing Research Lab** Jan 2024 – May 2024
IIIT Hyderabad

Conducted research on serverless function cold-start latency mitigation under fluctuating traffic patterns.

- Benchmarked OpenFaaS container provisioning times across varied Linux kernel cgroup configurations.
- Co-authored research paper draft accepted at National Student Cloud Computing Symposium.